

PineSORT: A Simple Online Real-time Tracking Framework for Drone Videos in Agriculture

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Abstract

We introduce *PineSORT*, a novel *Multiple Object Tracking (MOT)* system for drone-based agricultural monitoring, specifically tracking pineapples for yield estimation. Our approach tackles key challenges such as repetitive patterns, similar object appearances, low frame rates, and drone motion effects. *PineSORT* enhances the tracking accuracy with motion direction cost, camera motion compensation, a three-stage association strategy, and overlap management. To handle large displacements, we propose an ORB-based camera compensation technique that significantly improves the Association Accuracy (AssA). Evaluated via 5-fold cross-validation against *BoTSORT* and *AgriSORT*, *PineSORT* achieves statistically significant gains in our Identity-Switch Penalized IDF1 (ISP-IDF1) metric, along with gains in IDF1 (Identity F1 Score), HOTA (Higher Order Tracking Accuracy) and AssA. These results confirm its effectiveness in tracking **low-FPS drone footage**, making it a valuable tool for precision agriculture.

1. Introduction

Multiple Object Tracking (MOT) is a fundamental task in computer vision that involves detecting and tracking multiple objects across a sequence of video frames. The goal is to determine the object locations in each frame while maintaining their identities over time, even as they move, interact, or become occluded temporarily. **MOT** has broad applications in fields such as action recognition, pose estimation [28], surveillance [2], autonomous driving [40], sports analysis [9], and human-computer interaction. However, their application in agricultural environments, particularly for crop monitoring using drones, presents unique challenges that require specialized solutions.

Drones have become valuable tools in agriculture owing to their ability to capture high-resolution data over large areas. However, applying **MOT** in agricultural fields in-

troduces several challenges related to the characteristics of both the environment and objects being tracked. For instance, pineapples, similar to many agricultural products, exhibit high visual similarity and are arranged in repetitive patterns, making it difficult to distinguish individual instances accurately. Additionally, occlusions caused by leaves or other vegetation can partially or completely obscure fruits. These factors increase the likelihood of identity switches in tracking algorithms and reduce reliability.

Further complexity arises because of the motion of the drone. Even in cases where the drone moves in a single direction while the pineapples remain stationary, the objects in the image undergo continuous changes in position, scale, and perspective. Motion blur and changes in the frame rate can exacerbate tracking errors, particularly when the drone moves at high speeds, leading to inconsistent object associations.

Accurate **MOT** in agriculture is crucial for applications such as yield estimation, harvesting automation, and crop monitoring. However, several challenges hinder real-world implementation, including occlusions, dense object arrangements, and redundant detections. The latter occurs when multiple bounding boxes are generated for a single object, leading to tracking inconsistencies. Moreover, the sparse distribution of pineapples in fields, with large empty spaces between them, adds another layer of difficulty. Traditional tracking approaches often struggle with scale variations, occlusions, and high visual similarity among objects.

To tackle these challenges, it is crucial to develop advanced **MOT** methods that can effectively handle dynamic scenes caused by drone movement while maintaining real-time performance. These methods have to be robust against occlusions, redundant detections, and the repetitive patterns commonly found in pineapple fields. Overcoming these limitations will enhance **MOT** systems, improving agricultural monitoring and automation. This, in turn, enables more efficient and precise crop management. Additionally, ensuring accurate object counting is essential for reliable yield estimation and preventing multiple detections of the

same object.

In this work, we propose an **MOT** approach tailored for drone-based agricultural applications. Our method incorporates directional motion features, camera compensation using **Kalman Filter (KF)** state correction after prediction, and a three-stage object association based on **Intersection over Union (IoU)**. This approach is specifically designed for agricultural scenarios in which objects exhibit similar appearances, remain mostly stationary, and maintain relatively consistent sizes throughout tracking. Our key contributions are as follows:

- We introduce a novel **MOT** SORT-based tracker based on tracking-by-detection specifically designed for low-FPS drone video applications in agricultural environments.
- We propose the use of motion features and **Oriented FAST and Rotated BRIEF (ORB)** for camera compensation in stationary drone video footage.
- We conduct a statistical analysis to evaluate the performance of various tracking algorithms on a drone-based agricultural dataset.

2. Related work

2.1. Multiple Object Tracking

The **Simple Online and Realtime Tracking (SORT)** [6] family of algorithms has become a cornerstone in **MOT**, employing a tracking-by-detection paradigm that involves detection and association stages. In the detection stage, object detectors like YOLO [29], Faster R-CNN [31], or SSD [22] localize objects in each frame using bounding boxes, while the association stage links detections across frames using the *Hungarian algorithm* [20]. **SORT** variants have evolved to address various **MOT** challenges. DeepSORT [36] enhances SORT by incorporating appearance information, while ByteTrack [41] introduces a hierarchical data association strategy to leverage low-score detections. BoTSORT [1] integrates motion and appearance cues with camera-motion compensation, and StrongSORT [11] improves DeepSORT with tools like ALink for track recovery and GSI for handling temporary object losses. OCSORT [8] addresses *Kalman filter* limitations by incorporating observation-centric corrections, and DeepOCSORT [24] further integrates appearance cues. HybridSORT [39] leverages both strong and weak cues for improved performance in occlusion and clustering scenarios. Finally, SFSORT [26] introduces the Bounding Box Similarity Index, replacing the Kalman Filter. The **SORT** family is favored for its simplicity and adaptability in **MOT** tasks.

2.2. Multiple Object Tracking in Agriculture

There have been many recent advancement in precision agriculture [12–14, 16, 38]. Regarding multiple object tracking, it has focused on addressing challenges specific to

dynamic field environments. NTrack [27], a novel framework, utilizes dense optical flow and particle filtering to track cotton bolls, outperforming existing methods. Another approach employs fusion point cloud data and an improved **Graph Neural Network (GNN)** method for object tracking in farmlands, achieving high accuracy and precision [17]. An EKF-based multi-object tracking framework has been developed for mobile robots in precision agriculture, specifically for tracking table-grape bunches in vineyards [4]. AgriSORT, a simple online real-time tracking-by-detection pipeline, focuses on efficiency and flexibility for agricultural applications, particularly in challenging environments like table grape vineyards [34]. These methods enhance crop monitoring, yield prediction, and robotic harvesting but lack drone-based precision for yield estimation.

3. Proposed Method

PineSORT introduces a cutting-edge approach to **MOT** in agricultural drone footage, incorporating three key innovations that significantly enhance tracking performance. Our novel pipeline integrates (1) a camera compensation using **ORB** features (Fig. 1-Estimate Camera Motion), (2) a three-stage association process (Fig. 1-Third Association), and (3) a novel overlapping management (Fig. 1-Overlap Management based on IoU) that effectively handles challenging detection scenarios. These enhancements collectively address the unique challenges of drone-based tracking, such as low frame rates, repetitive patterns, and camera-induced motion distortions.

Our proposed pipeline works as follows: Upon receiving an input frame at time t , the system estimates camera motion compensation using **ORB** features between frames at times $t - 1$ and t (see details in Sec. 3.1). It then computes the affinity matrix transformation and correct the Kalman filter for each tracked object's state, executing the prediction step within the Kalman filter.

The object detector generates detection proposals categorized into low and high confidence detections. These detections are processed in the first association step, which employs a threshold τ based on the **IoU** metric to match object detections with tracked states. The second association step similarly utilizes **IoU** for matching. Unmatched detections and tracklets from the first two association steps are processed in a third association step (see Sec. 3.2), which identifies matches among the remaining unmatched detections and tracklets.

Matched tracklets from all three association steps are subsequently managed by the tracklet management system (Sec. 3.3). When a detection is absent, the system checks for overlaps using **IoU** with other tracklets to minimize redundancy. Furthermore, the frame is utilized to compute **ORB** features, which are matched using a brute-force matcher to refine the bounding box based on the drone's

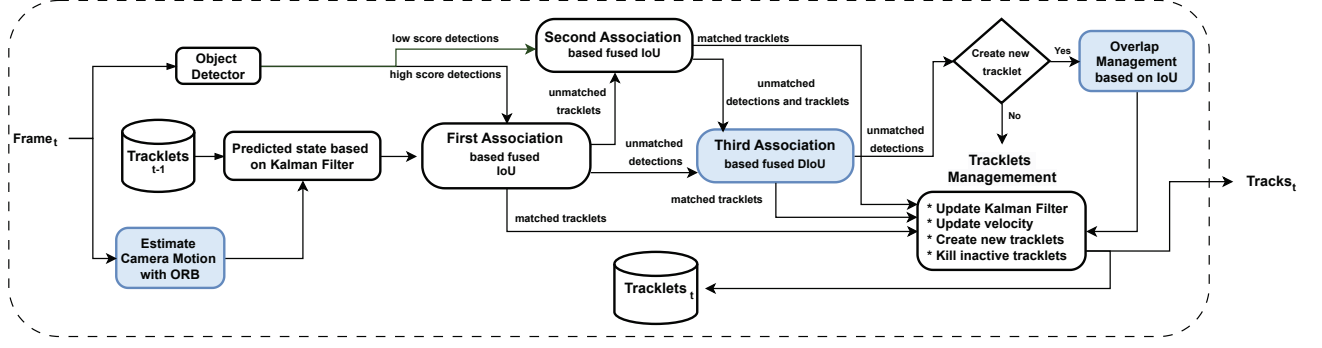


Figure 1. Overview of the PineSORT Pipeline: Our contributions are highlighted in blue. The pipeline begins with an object detector that generates both low- and high-score detections, which feed into the first two IoU-based association steps. Unmatched detections and tracklets then proceed to a third IoU-based association step. Matched tracklets undergo management to reduce object redundancy through IoU-based overlap checks. Additionally, ORB features are computed and matched using a brute-force matcher to refine bounding boxes in accordance with drone movement.

movement.

3.1. Camera Compensation with ORB

To address the impact of large displacements between consecutive frames caused by low frame rates (FPS), the system employs the **ORB** [33] algorithm, with the implementation of OpenCV [7]. This process consists of three main steps: keypoint detection, descriptor computation, and feature matching.

In the first step, **ORB** utilizes the FAST algorithm for keypoint detection, which identifies keypoints based on corner features within the image. For each detected keypoint, **ORB** computes a binary descriptor known as BRIEF, which is rotation-invariant and suitable for efficient matching.

The final step involves matching features. The **ORB** descriptors can be matched using a brute-force approach. Let I_t and I_{t-1} represent the current and previous frames, respectively. **ORB** detects keypoints $\mathbf{p}_t = \{p_{t,1}, p_{t,2}, \dots, p_{t,N}\}$ in I_t and $\mathbf{p}_{t-1} = \{p_{t-1,1}, p_{t-1,2}, \dots, p_{t-1,M}\}$ in I_{t-1} , along with their corresponding descriptors $\mathbf{d}_t$ and $\mathbf{d}_{t-1}$. Feature matching is performed by calculating the Euclidean distance $D(p_{t,i}, p_{t-1,j})$ between descriptors $\mathbf{d}_t$ and $\mathbf{d}_{t-1}$, thus identifying the set of matched keypoints $\mathcal{M} = \{(p_{t,i}, p_{t-1,j})\}$.

Using the matched keypoints $\mathcal{M}$, the system computes the affine transformation matrix $\mathbf{A}$, which maps the coordinates of keypoints from I_{t-1} to I_t . The affine transformation is represented as a 2×3 matrix:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \end{bmatrix},$$

where a_{11} , a_{12} , a_{21} , and a_{22} encode rotation, scaling, and shearing, while t_x and t_y represent translation in the x and y directions, respectively. The transformation is computed using **Random sample consensus (RANSAC)** [15]

from OpenCV [7].

The computed affine transformation matrix $\mathbf{A}$ is then applied to the state vector of the **KF** to update the predicted track positions. Let $\mathbf{x}_k = [x_k, y_k]^\top$ represent the x and y coordinates of the track's state vector at time k . The updated coordinates $\mathbf{x}'_k$ are computed as:

$$\mathbf{x}'_k = \begin{bmatrix} x'_k \\ y'_k \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}.$$

This transformation ensures that the **KF** predictions are aligned with the observed motion between frames, compensating for drone-camera motion and large object displacements (see the problem of misaligned prediction in Fig. 2). By integrating ORB-based feature matching and affine transformation into the tracking pipeline, the system achieves robust performance in low-FPS scenarios and dynamic agricultural environments.

3.2. Three-Stage Association

The proposed three-stage association module aims to associate tracklets with detections at varying confidence levels. In the first stage, tracklets are associated with high-confidence detections using a variant of the **IoU** metric. Let D_h represent the set of high-confidence detections with a threshold, and T denote the set of tracklets. A detection $d \in D_h$ is associated with a tracklet $t \in T$ if their **IoU** exceeds a predefined minimum of threshold t , expressed as $\text{IoU}(d, t) \geq t$.

In the second stage, tracklets are associated with low-confidence detections using a reduced threshold t' , where $t' < t$. Let D_l represent the set of low-confidence detections. Detections with confidence below t' are removed from the detection set D . The association condition for this stage is $\text{IoU}(d, t) \geq t' \quad \forall d \in D_l$.

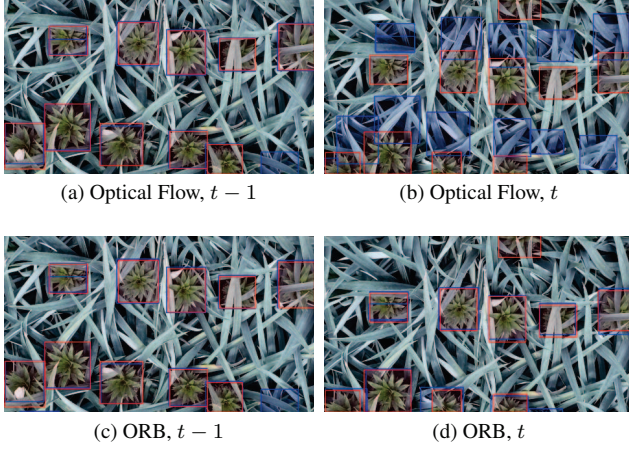


Figure 2. Comparison of Kalman filter predictions (blue) and object detector detections (red) across frames $t - 1$ and t . In Figures 2a and 2b, the analysis is conducted using Optical Flow, while Figures 2c and 2d utilize ORB. Specifically, Figure 2a illustrates alignment in frame $t - 1$ with Optical Flow, but Figure 2b reveals misalignment in frame t . Conversely, for ORB, Figure 2c shows alignment in frame $t - 1$, and Figure 2d confirms alignment in frame t .

In the final stage, partial objects are associated with tracklets using the **Distance IoU (DIOU)** metric. This metric evaluates both the overlap and the distance between the centers of bounding boxes, making it useful for aligning partial objects with **KF** predictions when they are exiting the scene. This step unifies the remaining detections that were not matched in the first two stages. Let D_u represent the set of unmatched detections, and T_u denote the set of tracklets not matched in the previous steps. The association condition for this stage is $\text{DIOU}(d, t) \geq t'' \quad \forall d \in D_u, t \in T_u$, where t'' is a threshold specific to the **DIOU** metric. The third stage is designed to match partial objects with tracklets that were not successfully associated in the first and second stages. For the association step in our experiments, we arbitrary utilized a threshold t , t and $t' \geq 0.30$ and $t'' \geq 0.10$.

3.3. Overlapping Management

The overlapping management mechanism computes the **IoU** between track bounding boxes B_{track} and detection bounding boxes B_{det} when creating a new track for detections not matched in the association steps. If the computed **IoU** exceeds a predefined threshold t , i.e., $\text{IoU}(B_{track}, B_{det}) > t$, redundant detections are removed to prevent ID switches. This mechanism is particularly critical in sparse agricultural environments, where overlapping objects are uncommon but can still occur due to detection errors, occlusions, or perspective effects. By filtering out redundant detections, the system enhances tracking accu-

racy and robustness, ensuring reliable performance in applications such as yield estimation and automated harvesting. The threshold t is empirically determined to optimize the trade-off between minimizing false positives and preserving valid detections, thereby maintaining the integrity of object trajectories.

4. Experiments

This section outlines the experimental setup, evaluation metrics, and benchmark results, comparing PineSORT to the baseline and analyzing its performance.

4.1. Dataset

The dataset includes 10 videos from a pineapple plantation from Upala Agricola [3] (the world's largest exporter of fresh pineapple), recorded with a DJI Mavic 3 drone at 1920×1080 resolution. Videos capture crop rows from vertical (90°) and oblique (45°) angles, lasting 7–35 seconds (183–882 frames each).

For **MOT**, we manually annotated the dataset using LabelStudio [35]. The annotations were converted into the **MOT** format [21], serving as the ground truth labels. The final annotations were exported at a frame rate of 25 **Frame per second (FPS)**.

4.2. Evaluation Metrics

We evaluate tracking performance using TrackEval [19], which provides comprehensive metrics. Overall performance is assessed with **Multiple Object Tracking Accuracy (MOTA)** [5], considering false positives, missed detections, and identity mismatches, along with **Identity F1 Score (IDF1)** [32] and **Higher Order Tracking Accuracy (HOTA)** [23], the geometric mean of **Detection Accuracy (DetA)** and **Association Accuracy (AssA)**.

Given the importance of preserving object identities for yield estimation, we emphasize **AssA**, which measures identity consistency, and **Identity Switches (IDSW)** [5], which counts identity switches. We also assess trajectory-level metrics: **Mostly tracked (MT)** (trajectories tracked for $\geq 80\%$ of their duration), **Mostly lost (ML)** (trajectories lost for $> 80\%$), and **Partially tracked (PT)** (objects tracked for 20–80% of their lifespan). Additional measures include **Identification Precision (IDP)** and **Identification Recall (IDR)** for identity-based analysis.

To account for identity preservation, we introduce the **Identity-Switch Penalized IDF1 (ISP-IDF1)** metric incorporating an exponential penalty based on **IDSW**. This normalization penalizes high ID switch counts while having a minimal effect when **IDSW** is low, for further analysis in the evaluation dataset:

$$\text{ISP-IDF1} = \text{IDF1} \times \left(1 - \frac{\text{IDSW} - \min(\text{IDSW})}{\max(\text{IDSW}) - \min(\text{IDSW})}\right)^k \quad (1)$$

where $k = 4$ was enough. This equation modifies the ranking of results by penalizing high IDSW values, emphasizing solutions with fewer identity switches. This aligns with reweighting functions used in optimization and machine learning to adjust scores adaptively [30].

In agricultural drone applications, maintaining consistent track identities is crucial for accurate object counting. Therefore, **AssA**, **IDF1**, **IDSW**, and **ISP-IDF1** are particularly relevant to our analysis.

4.3. Implementation details

We use YOLOv11n [18] as the detector for its speed-accuracy balance. The model was trained on the pineapple dataset using k-fold cross-validation, training on one video and evaluating on the rest. Training data included vertical, oblique, and combined camera orientations.

4.4. Trackers for comparison

We evaluate our approach against **state-of-the-art (SOTA)** SORT-based **MOT** methods, specifically SORT, ByteTrack, DeepOCSORT, StrongSORT, BoTSORT, and AgriSORT. In addition, to isolate the impact of motion-based tracking, we use BoTSORT without the **Re-Identification (ReID)** module in our analysis.

We also include HybridSORT and OCSORT, which improve robustness in occlusion and non-linear motion scenarios. Finally, we examine SFSORT, which replaces the **KF** with a custom cost function for better accuracy.

5. Results and Discussion

5.1. Camera Compensation Analysis

Table 1 compares the performance of **Scale-Invariant Feature Transform (SIFT)**, **ORB**, and Optical Flow for camera motion compensation. We conduct a 5-fold validation and report the mean and standard deviation of **IDF1**, **HOTA**, **AssA**, and **ISP-IDF1** across folds, along with the total **IDSW**. For **SIFT** and **ORB**, we compute descriptors and use a Brute-Force Matcher with k-nearest neighbors (k-NN) based on L2 distance to establish correspondences. For Optical Flow, we detect key points using the Shi-Tomasi method and track them with the Lucas-Kanade method. The affine transformation matrix is then computed using the **RANSAC** algorithm.

The results indicate that SIFT and ORB consistently achieve higher scores across key metrics. Specifically, **ORB** attains a mean **ISP-IDF1** score of approximately 0.63 with low variability (0.136), demonstrating robust tracking performance with fewer identity switches compared on using

Table 1. Performance comparison of camera compensation techniques using 5-fold cross-validation, reporting mean (μ) and standard deviation (σ) of IDF1, HOTA, AssA, ISP-IDF1, and total ID switches (IDSW).

Technique	IDF1 $\uparrow$		HOTA $\uparrow$		AssA $\uparrow$		IDSW $\downarrow$ Total	ISP-IDF1 $\uparrow$	
	μ	σ	μ	σ	μ	σ		μ	σ
ORB	0.836	0.064	0.636	0.056	0.712	0.039	131	0.630	0.136
Optical Flow	0.813	0.093	0.621	0.069	0.687	0.064	451	0.400	0.220
SIFT	0.835	0.064	0.636	0.056	0.712	0.039	129	0.634	0.137

optical flow. **SIFT** achieves similar performance across metrics, reflecting its effectiveness in maintaining object identity despite appearance variations. In bold the best mean is highlighted.

In contrast, Optical Flow exhibits lower mean **IDF1**, **HOTA**, **AssA**, and **ISP-IDF1** scores, along with a significantly higher identity switch count (451), highlighting difficulties in preserving object identities over time. Its lower **ISP-IDF1** score (0.4) suggests challenges in maintaining accurate object correspondences. Additionally, optical flow - in lower **FPS** - struggles to handle large object displacements, often resulting in track loss or identity switches during fast movements.

SIFT and **ORB** are best for tracking under low frame rates, while Optical Flow struggles due to higher identity switches and reduced consistency. Its assumption of small motion fails with large displacements [37].

5.2. PineSORT vs baselines trackers

Table 2 summarizes the performance of PineSORT, compared to baseline trackers. The evaluation uses a detector trained and tested on vertical, oblique, and combined videos via k-fold cross-validation. The table shows the mean (μ) and standard deviation (σ) for each tracker. The “Tracker” column lists the tracking algorithm, while the “Camera” column specifies the evaluation dataset (oblique or vertical). The “Training” column indicates the dataset used for training (combined, oblique, or vertical). Additionally, the table reports mean (μ), standard deviation (σ), and total sums for PT, MT, ML, and IDSW.

To identify the suitable statistical method for comparing the ten tracker groups in our experiment, we evaluated the assumptions of normality and homoscedasticity using R software. The analysis revealed that none of the tracker groups exhibited a normal distribution or satisfied the homoscedasticity assumption. As a result, we applied the Kruskal-Wallis test [25], followed by a post-hoc Dunn test [10], for our statistical analysis.

The Kruskal-Wallis test produced a p-value of $< 2.2 \times 10^{-16}$ based on the **ISP-IDF1** metric, which is significantly lower than the 0.05 threshold. Consequently, we reject the null hypothesis, confirming that the performance distribution of at least one tracker differs from the others. To iden-

Table 2. Performance comparison of tracking algorithms on vertical and oblique datasets. Object detector models were trained on vertical, oblique, or combined data using K-fold cross-validation. Metrics (MOTA, HOTA, IDF1, AssA, IDP, IDR, ISP-IDF1) are reported as mean (μ) $\pm$ standard deviation (σ); PT, MT, ML, and IDSW are shown as total counts across folds. Bold and underlined values indicate the best performance per metric.

Tracker	Camera	Training	MOTA $\uparrow$		HOTA $\uparrow$		IDF1 $\uparrow$		AssA $\uparrow$		IDP $\uparrow$		IDR $\uparrow$		PT Total	MT $\uparrow$ Total	ML $\downarrow$ Total	IDSW $\downarrow$ Total	ISP-IDF1 $\uparrow$	
			μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ					μ	σ
AgriSORT	oblique	combined	-1.068	0.861	0.256	0.084	0.323	0.135	0.309	0.103	0.243	0.110	0.500	0.170	3323	1731	974	6082	0.109	0.103
		oblique	-1.261	1.379	0.222	0.104	0.273	0.172	0.267	0.119	0.222	0.139	0.390	0.230	2929	1207	1892	5411	0.099	0.090
		vertical	-1.168	0.697	0.181	0.049	0.199	0.087	0.224	0.063	0.163	0.072	0.281	0.130	3968	605	2962	7205	0.063	0.052
	vertical	combined	-2.205	1.622	0.203	0.081	0.229	0.154	0.272	0.097	0.162	0.118	0.405	0.204	3586	775	1559	4443	0.107	0.108
		oblique	-2.725	2.623	0.179	0.077	0.186	0.140	0.249	0.079	0.135	0.111	0.318	0.178	3936	579	2885	5164	0.076	0.074
		vertical	-1.992	1.338	0.205	0.079	0.228	0.153	0.270	0.090	0.166	0.121	0.381	0.191	3452	650	1818	4109	0.108	0.099
BoTSORT	oblique	combined	0.591	0.173	0.576	0.079	0.768	0.110	0.643	0.060	0.785	0.088	<u>0.755</u>	0.132	1647	<u>3847</u>	534	1079	0.610	0.142
		oblique	0.369	0.398	<u>0.477</u>	0.165	0.624	0.245	<u>0.569</u>	0.130	0.708	0.224	<u>0.588</u>	0.259	1959	<u>2542</u>	1527	949	0.511	0.228
		vertical	0.263	0.173	<u>0.382</u>	0.086	<u>0.522</u>	0.144	<u>0.484</u>	0.064	0.678	0.119	<u>0.441</u>	0.147	3584	<u>1448</u>	2503	1426	0.406	0.131
	vertical	combined	0.664	0.216	0.624	0.077	0.811	0.100	0.682	0.073	0.845	0.113	0.785	0.097	1556	4078	286	795	0.686	0.130
		oblique	0.270	0.687	0.477	0.139	0.636	0.226	0.583	0.093	0.690	0.252	0.599	0.207	2711	3124	<u>1565</u>	1037	0.529	0.200
		vertical	0.593	0.304	0.590	0.085	0.781	0.122	0.663	0.069	0.823	0.151	0.750	0.098	1778	3711	431	728	0.671	0.137
ByteTrack	oblique	combined	0.381	0.153	0.437	0.084	0.595	0.132	0.543	0.055	0.814	0.111	0.484	0.137	2719	1242	2067	<u>237</u>	0.566	0.134
		oblique	0.271	0.207	0.320	0.173	0.426	0.259	0.450	0.150	<u>0.792</u>	0.153	0.327	0.222	1969	706	3353	<u>169</u>	0.410	0.253
		vertical	0.127	0.088	0.210	0.083	0.254	0.136	0.373	0.087	0.761	0.135	0.161	0.096	1777	144	5614	<u>253</u>	0.243	0.132
	vertical	combined	0.393	0.109	0.458	0.059	0.622	0.090	0.529	0.071	0.747	0.077	0.535	0.099	3193	1467	1260	910	0.518	0.144
		oblique	0.268	0.162	0.321	0.136	0.431	0.219	0.472	0.124	<u>0.801</u>	0.198	0.324	0.184	2836	738	3826	529	0.388	0.207
		vertical	0.398	0.135	0.440	0.063	0.610	0.106	0.520	0.073	0.784	0.105	0.504	0.106	3106	1290	1524	777	0.525	0.152
DeepOCSORT	oblique	combined	0.563	0.150	0.523	0.088	0.718	0.116	0.581	0.070	0.846	0.070	0.630	0.135	3035	2102	891	1313	0.545	0.154
		oblique	0.380	0.286	0.412	0.172	0.555	0.251	0.490	0.148	0.769	0.210	0.464	0.243	2800	1187	2041	1162	0.429	0.220
		vertical	0.227	0.126	0.294	0.085	0.413	0.144	0.387	0.070	0.751	0.114	0.298	0.122	3734	241	3560	1485	0.315	0.120
	vertical	combined	0.610	0.113	0.556	0.073	0.763	0.075	0.609	0.067	0.909	0.062	0.662	0.094	4118	1343	459	453	0.689	0.064
		oblique	0.360	0.318	0.406	0.130	0.574	0.214	0.498	0.095	0.782	0.232	0.463	0.192	4654	602	2144	721	0.505	0.189
		vertical	0.574	0.118	0.526	0.071	0.736	0.080	0.589	0.063	0.903	0.064	0.625	0.093	4217	1062	641	425	0.669	0.064
HybridSORT	oblique	combined	0.560	0.157	0.521	0.091	0.713	0.122	0.578	0.077	0.846	0.071	0.624	0.144	2952	2120	956	1367	0.541	0.179
		oblique	0.382	0.283	0.406	0.180	0.546	0.261	0.480	0.163	0.763	0.211	0.455	0.254	2643	1236	2149	1233	0.428	0.246
		vertical	0.220	0.129	0.282	0.093	0.390	0.155	0.370	0.087	0.732	0.115	0.279	0.132	3414	283	3838	1777	0.290	0.145
	vertical	combined	0.537	0.159	0.502	0.115	0.669	0.146	0.543	0.125	0.842	0.088	0.566	0.172	3796	1427	697	1861	0.477	0.239
		oblique	0.318	0.288	0.365	0.142	0.499	0.222	0.448	0.121	0.734	0.218	0.392	0.203	4142	703	2555	1799	0.367	0.213
		vertical	0.508	0.150	0.477	0.107	0.650	0.137	0.528	0.117	0.841	0.091	0.539	0.153	3932	1140	848	1673	0.475	0.218
OCSORT	oblique	combined	0.552	0.153	0.507	0.095	0.694	0.126	0.562	0.079	0.853	0.066	0.594	0.146	3070	1876	1082	1548	0.505	0.173
		oblique	0.380	0.264	0.389	0.184	0.525	0.264	0.466	0.162	0.784	0.192	0.427	0.250	2590	1086	2352	1183	0.407	0.234
		vertical	0.208	0.122	0.263	0.092	0.362	0.154	0.354	0.084	0.753	0.118	0.251	0.125	3183	199	4153	1669	0.270	0.135
	vertical	combined	0.531	0.155	0.495	0.114	0.661	0.145	0.538	0.122	0.852	0.078	0.552	0.169	3823	1329	768	1914	0.463	0.229
		oblique	0.334	0.243	0.358	0.143	0.491	0.224	0.449	0.119	0.769	0.206	0.377	0.201	4026	621	2753	1573	0.370	0.204
		vertical	0.508	0.137	0.472	0.104	0.646	0.131	0.526	0.110	0.862	0.065	0.525	0.148	3895	1074	951	1625	0.471	0.202
SFSORT	oblique	combined	0.388	0.163	0.390	0.111	0.490	0.167	0.492	0.078	<u>0.860</u>	0.081	0.361	0.159	2110	1054	2864	1485	0.351	0.144
		oblique	0.254	0.195	0.271	0.169	0.333	0.229	0.366	0.177	0.778	0.158	0.235	0.183	1653	560	3815	1086	0.248	0.175
		vertical	0.112	0.072	0.154	0.074	0.175	0.106	0.263	0.095	<u>0.762</u>	0.098	0.103	0.067	1398	72	6065	900	0.145	0.088
	vertical	combined	0.593	0.096	0.515	0.097	0.634	0.132	0.512	0.139	0.791	0.128	0.533	0.131	1724	2963	1233	4414	0.290	0.241
		oblique	0.306	0.204	0.302	0.155	0.369	0.221	0.389	0.140	0.754	0.181	0.272	0.187	2065	1341	3994	2957	0.207	0.167
		vertical	0.523	0.086	0.463	0.084	0.579	0.121	0.482	0.128	0.786	0.129	0.462	0.114	1998	2337	1585	3873	0.289	0.216
SORT	oblique	combined	0.528	0.146	0.488	0.080	0.693	0.112	0.539	0.069	0.799	0.080	0.617	0.129	3256	1897	875	1458	0.508	0.144
		oblique	0.351	0.299	0.395	0.156	0.545	0.238	0.462	0.137	0.717	0.217	0.465	0.233	2914	1169	1945	1464	0.400	0.209
		vertical	0.215	0.128	0.290	0.080	0.409	0.138	0.372	0.071	0.690	0.129	0.302	0.119	3829	263	3443	1812	0.295	0.114
	vertical	combined	0.535	0.131	0.483	0.073	0.714	0.086	0.537	0.066	0.861	0.079	0.618	0.108	4388	965	567	554	0.631	0.075
		oblique	0.292	0.352	0.369	0.115	0.545	0.202	0.453	0.087	0.722	0.230	0.444	0.181	4810	442	2148	906	0.465	0.178
		vertical	0.499	0.149	0.463	0.074	0.690	0.091	0.524	0.064	0.848	0.095	0.587	0.099	4446	736	738	541	0.611	0.076
StrongSORT	oblique	combined	0.517	0.165	0.512	0.084	0.690	0.118	0.558	0.078	0.750	0.092	0.644	0.138	2666	2596	766	2942	0.371	0.176
		oblique	0.320	0.355	0.412	0.161	0.546	0.237	0.473	0.145	0.669	0.219	0.488	0.242	2627	1669	1732	2732	0.305	0.183
		vertical	0.190	0.143	0.305	0.077	0.416	0.126	0.376	0.067	0.622	0.114	0.326	0.120	3935	483	3117	3660	0.215	0.103
	vertical	combined	0.482	0.192	0.465	0.111	0.615	0.142	0.484	0.124	0.731	0.109	0.542	0.168	3810	1624	486	3883	0.297	0.195
		oblique	0.171	0.538	0.350	0.128	0.470	0.199	0.405	0.113	0.594	0.222	0.399	0.185	4356	1045	1999	4117	0.230	0.155
		vertical	0.430	0.223	0.436	0.108	0.589	0.143	0.465	0.117	0.710	0.127	0.512	0.157	3866	1393	661	3706	0.289	0.184
PineSORT (ours)	oblique	combined	<u>0.615</u>	0.162	<u>0.587</u>	0.082	<u>0.784</u>	0.111	<u>0.651</u>	0.062	0.822	0.075	<u>0.755</u>	0.138	1793	3745	<u>490</u>	491	<u>0.706</u>	0.121
		oblique	<u>0.404</u>	0.378	<u>0.477</u>	0.176	<u>0.630</u>	0.254	0.557	0.152	0.752	0.226	0.574	0.265	2197	2332	<u>1499</u>	537	<u>0.570</u>	0.258
		vertical	<u>0.272</u>	0.150	0.366	0.092	0.507	0.150	0.457	0.074	0.717	0.105	0.411	0.149	3952	1083	<u>25</u>			

tify specific differences among the tracker groups, we performed a post-hoc Dunn test. Notably, when comparing BoTSORT (the best baseline tracker) and PineSORT using the **ISP-IDF1** metric, the test yielded a p-value of 0.0244, resulting in the rejection of the null hypothesis H_0 . This indicates a statistically significant difference between these two trackers, with PineSORT demonstrating superior performance in maintaining ID integrity in tracking.

Among the evaluated trackers, AgriSORT exhibited the least favorable performance metrics, characterized by a notably high **IDSW** across various combinations of camera evaluation settings and training protocols. This high **IDSW** was the highest recorded among all trackers, coupled with low **ISP-IDF1** scores. However, AgriSORT demonstrated a high PT, indicating effective tracking within 20%-80% of the object lifespan. This suboptimal performance can be attributed to its reliance on the Lucas-Kanade method for sparse optical flow to estimate camera motion. Additionally, AgriSORT replaces the Kalman filter's transition matrix with an affine transformation between consecutive frames $t-1$ and t during the state prediction phase. The efficacy of this filtering approach is highly contingent upon the congruence of the affine matrix with the underlying system dynamics. Discrepancies in the affine matrix can lead to significant degradation of track predictions, particularly in low **FPS** scenarios, where increased matching errors and challenges in accurately capturing motion components—such as translation, rotation, and scaling—culminate in heightened estimation errors. Furthermore, the optical flow technique operates under the assumption of small camera displacements, which limits its effectiveness in scenarios involving substantial motion.

As observed in Table 2, SFSORT exhibits a considerable number of identity switches, resulting in a lower **IDR** compared to other trackers. This indicates its inadequacy in accurately maintaining object identities across frames in accordance with the ground truth. These performance metrics highlight the limitations of its cost index function, which relies solely on object bounding box characteristics and lacks integration of a Kalman filter for motion prediction. The absence of a Kalman filter compromises temporal consistency, leading to an overreliance on shape similarity, distance, and bounding box overlap. When an object is only partially visible upon entering or exiting the frame, its size may be underestimated, causing a decline in shape similarity scores as the object becomes fully visible in subsequent frames. Moreover, larger displacements between frames at **FPS** further increase the likelihood of identity switches, as they contribute to a decrease in bounding box overlap.

BoTSORT demonstrates a lower ID switch rate compared to most other tracking methods, with the exception of PineSORT. However, it still experiences some ID switches due to its reliance on sparse optical flow for feature track-

ing and RANSAC for affine matrix estimation. While this approach aims to accurately estimate background motion, the inherent assumptions of optical flow may not adequately capture large displacements between frames. We anticipated that StrongSORT and DeepOCSORT, which depend on feature embeddings to enhance object association between tracks and predictions, would perform poorly since they were not trained on our dataset for feature embeddings comparison. StrongSORT, which relies heavily on appearance features, exhibits a high frequency of ID switches in our scenario, particularly when tracking homogeneous objects, leading to mismatches and inaccuracies, especially at low FPS with significant object displacement. In contrast, DeepOCSORT demonstrates improved performance over StrongSORT, likely due to the introduction of Camera Motion Compensation using optical flow and RANSAC, resulting in a substantial difference in the **ISP-IDF1** metric.

There is not a statistical difference between the trackers HybridSORT and OCSORT indicating similar performance for our benchmark in agricultural applications, as both tracking algorithms are predicated on the same SORT framework. Given the unidirectional trajectory of the drone, the implementation of camera compensation is critical for accurately capturing the dynamics of this movement. In instances where the object detector fails to identify a target in the subsequent frame, it may inadvertently assign new identifiers to newly detected objects in the next frame, while neglecting to update the states of the previously tracked objects. The tracking algorithms that do not integrate this feature include SORT, HybriSORT, OC-SORT, ByteTrack, and StrongSORT. Conversely, AgriSORT, BoTSORT, and PineSORT uniquely incorporate camera compensation techniques to facilitate the updating and correction of object states based on camera movement.

The analysis of tracking performance across various configurations of training data reveals that trackers utilizing a combined dataset of vertical and oblique orientations demonstrate superior performance. This enhancement can be attributed to the increased volume of training data, which allows the model to better capture the inherent variability present in agricultural environments. These environments are often characterized by clutter, diverse variations, and the unique challenges posed by pineapples obscured by foliage.

Notably, training models with vertical data while evaluating them under the same conditions leads to enhanced performance. This observation suggests that recording agricultural fields using a drone camera in a vertical orientation optimally facilitates the detection and tracking of objects, resulting in reduced variability and improved distinguishability of the target objects. Conversely, models trained and evaluated using oblique data exhibit lower performance levels. The diverse perspectives captured by the drone when operating in an oblique orientation introduce increased vari-

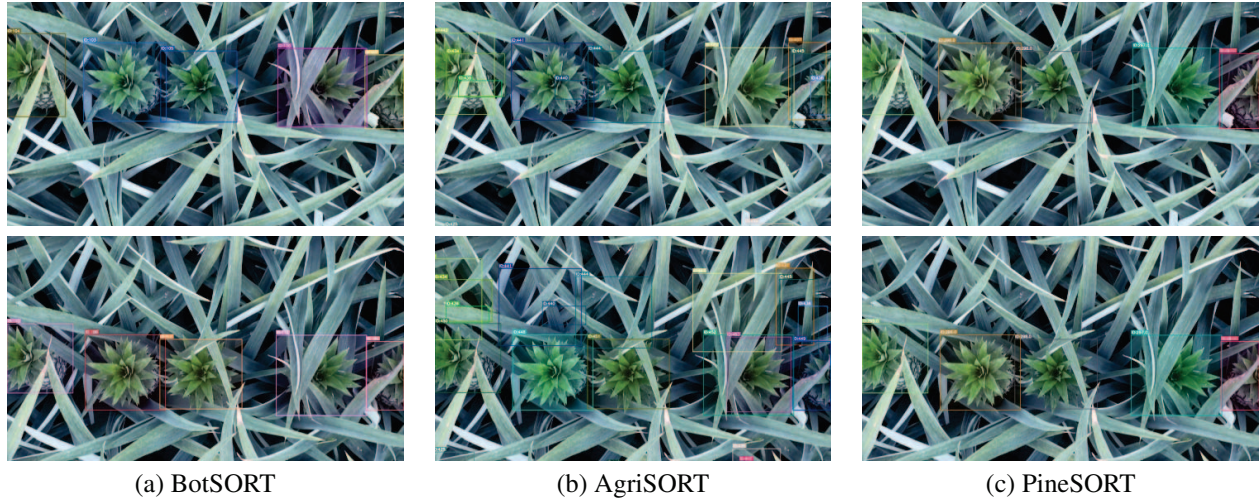


Figure 3. Qualitative comparison of motion-based tracking algorithms—BotSORT, AgriSORT, and PineSORT (*ours*)—across consecutive time frames. The upper panels display the tracking results at time frame $t - 1$, while the lower panels show the corresponding results at time frame t . Bounding boxes represent tracklets, with consistent colors indicating the same object ID and different colors denoting distinct IDs. As shown, PineSORT shows the most consistent results in localization and identification in consecutive frames.

ability, necessitating additional training data to adequately account for the different appearances of the pineapples.

Furthermore, it is essential to highlight that training a model on data captured from a vertical camera orientation and subsequently evaluating it with an oblique dataset results in suboptimal performance. Similarly, the reverse scenario, where a model trained on oblique data is evaluated on vertical data, also yields poor performance. These findings underscore the limitations of supervised learning concerning generalization across different perspectives. Such limitations emphasize the need for a robust training strategy with wider variety of perspectives to enhance the model’s adaptability and effectiveness in real-world applications.

5.3. Motion-based Method Qualitative Analysis

In our qualitative comparative analysis of motion tracker-based methods, we evaluate our PineSORT algorithm alongside BoTSORT and AgriSORT. As illustrated in Fig. 3, PineSORT consistently maintains object IDs across consecutive frames $t - 1$ and t . In contrast, BoTSORT experiences frequent ID switches due to significant displacements between tracked and detected objects, resulting in the assignment of new IDs in frame t and compromising identity consistency. This inconsistency is primarily attributed to BoTSORT’s reliance on the Lukas-Kanade optical flow method for estimating motion between frames.

Similarly, AgriSORT encounters challenges with unmatched tracked objects resulting from large displacements, leading to the generation of new IDs, akin to the issues faced by BoTSORT. AgriSORT employs an affinity matrix derived from Lukas-Kanade optical flow, which substitutes

the transition matrix F in the Kalman filter’s prediction step from frame $t - 1$ to frame t . This method does not remove unmatched tracks during track management, leading to increased identity switches and adversely affecting the consistent identification of each object. Consequently, these limitations hinder the performance of agricultural video drone counting analyses.

6. Conclusion

In this paper, we introduced PineSORT, a novel MOT tracker designed for drone videos in agricultural settings. The framework integrates key components such as a three-stage association process, motion direction cost, camera compensation, and overlap management, significantly improving tracking performance, especially in dynamic conditions with rapid object movement and low frame rates. We improved bounding box predictions and reduced identity switches in dense environments by integrating a KF with a constant-velocity model and expanding the state vector to include width and height.

Tested on drone footage from a pineapple plantation case, PineSORT leverages ORB for effective camera compensation in low FPS drone footage, outperforming traditional optical flow method. PineSORT consistently outperformed baseline trackers, including BoTSORT, in terms of IDF1 and AssA scores, demonstrating its robustness in agricultural object tracking. Overall, PineSORT provides a reliable solution for agricultural tracking, laying the groundwork for future developments in yield estimation and automated harvesting, with a focus on scalability and improving performance in more challenging environments.

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